

LAB GUIDELINES

Lab 7 — Partition and Volume Types, File Systems

Maps to CompTIA Server+ objective **2.1** — Partition and volume types (GPT vs. MBR, dynamic disk, LVM) and File system types (ext4, NTFS, VMFS, ReFS, ZFS).

Run all commands on <https://killercode.com/playgrounds/scenario/ubuntu>

Required software (free, apt): parted, gdisk, e2fsprogs, xfsprogs, zfsutils-linux, ntfs-3g

```
apt update && apt install -y parted gdisk e2fsprogs xfsprogs zfsutils-linux ntfs-3g
```

Step 1 — Create a "disk" you can partition

```
truncate -s 4G /srv/disk.img
losetup -P /dev/loop10 /srv/disk.img
```

Step 2 — MBR (Master Boot Record) partition table

```
parted -s /dev/loop10 mklabel msdos
parted -s /dev/loop10 mkpart primary ext4 1MiB 500MiB
parted -s /dev/loop10 mkpart primary ext4 500MiB 1000MiB
parted /dev/loop10 print
```

MBR limits: max **2 TiB** disk, max **4 primary** partitions (or 3 + 1 extended). Legacy BIOS booting.

Step 3 — Switch to GPT (GUID Partition Table)

```
parted -s /dev/loop10 mklabel gpt
parted -s /dev/loop10 mkpart data1 ext4 1MiB 1GiB
parted -s /dev/loop10 mkpart data2 xfs 1GiB 2GiB
parted -s /dev/loop10 mkpart data3 ext4 2GiB 3GiB
parted /dev/loop10 print
gdisk -l /dev/loop10
```

GPT: max **9.4 ZB** disk, **128 partitions** by default, redundant header (backup at end of disk), required by UEFI booting.

Step 4 — Format with each file system

```
mkfs.ext4 /dev/loop10p1 # ext4 - Linux default
mkfs.xfs /dev/loop10p2 # XFS - large file workloads
zpool create -f tank /dev/loop10p3 # ZFS - pooled, checksumming, snapshots
zfs list
```

Step 5 — File system comparison (Server+ exam mapping)

FS	Native OS	Use case	Notes
ext4	Linux	General-purpose Linux servers	Journalled, mature, simple
XFS	Linux (RHEL default)	Large files, parallel I/O, big mounts	Online grow only (no shrink)
NTFS	Windows	Windows Server volumes, NTFS ACLs	Read/write on Linux via `ntfs-3g`
ReFS	Windows Server	Storage Spaces, large data integrity workloads	Checksumming, no boot
VMFS	VMware ESXi	Datastore for `.vmdk` files on ESXi hosts	Clustered, multi-host concurrent access

FS	Native OS	Use case	Notes
ZFS	Linux/FreeBSD/Solaris	Pooled storage with checksums, snapshots, send/recv	Volume manager + FS in one

Step 6 — Dynamic disk vs. LVM

`dynamic disk` is the Windows concept (spanned/striped/mirrored volumes in Disk Management). The Linux equivalent is **LVM** (Lab 3). Both let you grow, span, and mirror across multiple physical disks without changing the partition table.

`pvs && vgs && lvs` # see Lab 3 – LVM is the Server+ answer on Linux

Step 7 — Mounting, persistence with `/etc/fstab`, and UUIDs

```
mkdir -p /mnt/p1 /mnt/p2
mount /dev/loop10p1 /mnt/p1
blkid /dev/loop10p1
# Persist by UUID (not /dev/loopN which changes between reboots):
UUID=$(blkid -s UUID -o value /dev/loop10p1)
echo "UUID=$UUID /mnt/p1 ext4 defaults,noatime 0 2" | tee -a /etc/fstab
mount -a
```

Step 8 — Convert MBR → GPT non-destructively

```
losetup -d /dev/loop10 && losetup -P /dev/loop10 /srv/disk.img
parted -s /dev/loop10 mklabel msdos
parted -s /dev/loop10 mkpart primary ext4 1MiB 1GiB
gdisk /dev/loop10 <<'EOF'
r
g
w
y
EOF
gdisk -l /dev/loop10 | head
```

The `gdisk` recovery menu can convert MBR → GPT in place (after a full backup!).

Free online tools

- ****GNU parted manual**** — <https://www.gnu.org/software/parted/manual/>
- ****ZFS on Linux**** — <https://openzfs.github.io/openzfs-docs/>
- ****NTFS-3G**** — <https://www.tuxera.com/community/open-source-ntfs-3g/>

What you learned

- GPT vs. MBR limits and when to choose each.
- The five Server+ filesystems (ext4, NTFS, VMFS, ReFS, ZFS) and where each one lives.
- LVM as the Linux answer to Windows dynamic disks.
- UUID-based `/etc/fstab` for survivable mounts.
- Non-destructive MBR → GPT migration with `gdisk`.